Community Detection In Networks: Evaluating SCORE Through Simulation & Real-World Applications

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Abstract

Community detection aims to infer groups from network connections when node labels are unknown. Standard spectral clustering can be affected by degree heterogeneity: nodes that are in the same community may have a very different numbers of connections. This project studies SCORE, a method introduced by Jiashun Jin (2015), for networks modeled by the degree-corrected stochastic block model. SCORE computes leading eigenvectors of the adjacency matrix and represents each node by ratios of eigenvector entries. Because degree effects can appear as a shared multiplicative factor across these entries, taking ratios can reduce their influence before clustering. The project examines SCORE's performance on a political blogs network and simulated networks, comparing it with ordinary and normalized spectral clustering.

Introduction

Many datasets are naturally represented as networks. A network consists of nodes, which represent the entities being studied, and edges, which represent relationships between pairs of entities. In community detection, the task is to infer groups from the pattern of connections: nodes in the same group are expected to connect more often to one another than to nodes in other groups. The labels are unknown during estimation, so the method must recover the groups from the network itself.

The political blogosphere surrounding the 2004 United States presidential election gives a great example. Each node represents a political blog, and an edge represents a hyperlink between two blogs. Following Jin (2015), hyperlink direction is ignored, so the network is treated as undirected. The data contain two recognizable communities associated with liberal and conservative political orientations. Those orientations are useful as reference labels for evaluating a clustering method, but the method estimates groups from hyperlinks rather than being given the labels.

The challenge is that blogs differ in popularity. A widely read blog may receive or make many links, while a less prominent blog may have few. This difference in degree can occur even when two blogs belong to the same political community. A basic stochastic block model (SBM) represents connection patterns through community membership: blogs assigned to the same community share the same expected connection profile. This is a useful simplification, but it does not represent the large within-community differences in popularity that can occur in real networks. When the model is a poor fit, conventional clustering may group nodes partly by their overall number of connections rather than by the community pattern we want to identify.

The degree-corrected block model (DCBM) refines this structure by allowing each node to have its own degree tendency. It can therefore represent both the community-level pattern of connections and the fact that some nodes are hubs. Estimating every node's degree tendency directly and separating it from community structure can be difficult. SCORE offers a spectral approach: it computes leading eigenvectors of the adjacency matrix and clusters ratios of their entries. The assumption is that degree heterogeneity affects these eigenvectors in a shared way, so taking ratios can largely cancel the popularity effect and leave features that more clearly distinguish communities. This cancellation is approximate, but it gives SCORE a way to address degree variation without first estimating a separate degree parameter for every node.

Compared to ordinary spectral clustering, which clusters the eigenvector coordinates directly, SCORE's ratios are designed to be less sensitive to degree heterogeneity. Normalized spectral clustering is also another common strategy: it adjusts the adjacency matrix using node degrees before computing eigenvectors. Comparing these methods on the same network helps show the practical difference between using raw eigenvectors, degree normalization, and eigenvector ratios. Jin's paper develops theoretical conditions for reliable community recovery and demonstrates SCORE on the political blogs network. In this project, the blog labels provide a benchmark for comparing SCORE with ordinary and normalized spectral clustering. Synthetic DCBM experiments complement the real-data example by varying degree heterogeneity and community separation under controlled conditions. Together, these analyses ask when accounting for node popularity improves community detection and how SCORE performs relative to traditional spectral alternatives.

Methods

Network and degree-corrected block model

Let an undirected network have n nodes and adjacency matrix $A \in \{0,1\}^{n\times n}$. For distinct nodes i and j, $A_{ij} = 1$ if there is an edge and $A_{ij} = 0$ otherwise. The adjacency matrix is symmetric and has a zero diagonal when self-loops are excluded. Let $g_i \in \{1,\dots,K\}$ be the unknown community of node i, and let $\theta_i > 0$ describe its relative tendency to form edges. The DCBM assumes

$$\Pr(A_{ij} = 1 \mid g, \theta) = \theta_i \theta_j P_{g_i g_j}, \qquad i < j,$$

where P is a symmetric $K \times K$ matrix of baseline connection probabilities between communities. Conditional on the parameters, edge indicators for distinct unordered pairs are independent Bernoulli variables. The values of (P) describe the community connection pattern; the node parameters θ_i allow nodes in the same community to have different expected degrees. All probabilities must be in ([0,1]).

The community structure is low-dimensional when K is small relative to n: connection patterns are represented by the $K \times K$ matrix P, together with one community label per node. The labels are the primary quantities of interest. The degree parameters account for popularity differences that could otherwise obscure those labels.

SCORE estimation and intuition

Eigenvectors summarize large-scale patterns in a matrix. When a network has stronger connections within communities than between them, leading eigenvectors of its adjacency matrix can give nodes with similar community roles similar coordinates. Clustering those coordinates can reveal groups.

Degree heterogeneity can distort these coordinates. A node's eigenvector entries may reflect both its community and its overall tendency to form edges. Under the DCBM, the node-specific degree effect can appear as a shared multiplicative factor across eigenvector entries. SCORE uses ratios to approximately cancel this common factor.

Let $\hat{\eta}_1, \ldots, \hat{\eta}_K$ be the eigenvectors of the observed adjacency matrix A associated with the K largest eigenvalues in absolute value. SCORE constructs:

$$\widehat{R}(i,k) = \frac{\widehat{\eta}_{k+1}(i)}{\widehat{\eta}_1(i)}, \qquad i = 1, \ldots, n, \quad k = 1, \ldots, K-1.$$

Here, k indexes the ratio coordinate, which is the eigenvector being divided by the first eigenvector. i identifies the node, and $k = 1, \ldots, K-1$ selects one of the $K-1$ ratios for that node. The ratio uses eigenvector $\widehat{\eta}_{k+1}$ divided by $\widehat{\eta}_1$. So for example, with $K = 2$, there is one ratio per node: $\widehat{\eta}_2(i)/\widehat{\eta}_1(i)$. With $K = 3$, there are two ratios per node.

Each row of $\widehat{R}$ is a feature vector for one node. For two communities it has a single ratio coordinate. SCORE clips extreme coordinates to limit the effect of near-zero denominators, then applies K-means to the ratio vectors. The cancellation is approximate for an observed, noisy network; it does not guarantee perfect recovery.

Comparison methods and evaluation

The analysis compares three methods:

1. **SCORE:** K-means on ratios of leading eigenvectors of A.
2. **Ordinary spectral clustering (oPCA):** K-means on leading eigenvectors of A directly.
3. **Normalized spectral clustering (nPCA):** K-means on leading eigenvectors of $D^{-1/2}AD^{-1/2}$, where D is the diagonal matrix of node degrees.

The methods differ in how degree variation enters the clustering features. oPCA uses the raw adjacency eigenvectors, while nPCA normalizes by degree, and SCORE forms ratios intended to cancel a shared degree effect.

Because cluster labels are arbitrary, the estimated label 1 is not inherently liberal or conservative. Misclassification rate is therefore minimized over the two possible label mappings. The adjusted Rand index (ARI) is a complementary measure of agreement between two partitions, adjusted for chance. Larger values do indicate stronger agreement, with 1 representing a perfect match.

Numerical Experiments

Political blogs network and processing

The real-data analysis uses the `blogosphere` dataset, which is distributed with the R package `gsbm`. It contains 1,222 blogs, 16,714 undirected connections, blog names, and manually assigned political orientations. The adjacency matrix is symmetrized and self-loops are removed. The labels are held out during clustering and used only for evaluation. The code records the random seed and calculates misclassification rate and ARI for each method.

Preliminary real-data results

The analysis file `11_RMD_PoliticalBlogsReplication.Rmd` contains the code that fits and evaluates the methods. The current reported results are:

Method	Misclassified blogs	Error rate	ARI
SCORE	58	4.7%	0.819
Ordinary spectral clustering (oPCA)	437	35.8%	0.080
Normalized spectral clustering (nPCA)	590	48.3%	0.000

SCORE has the strongest agreement with the reference political labels in this run: it makes 58 errors and has an ARI of 0.819. This matches Jin's reported SCORE error count of 58/1,222 and the ARI recorded in the project materials. oPCA makes 437 errors (35.8% misclassification rate) and has ARI 0.080, indicating weak agreement. nPCA makes 590 errors (48.3%) and has ARI 0.000 to the displayed precision, indicating little adjusted agreement with the reference partition in this run. These results suggest that SCORE is the most effective of the three methods on this particular real network; they do not establish that it will always outperform degree-normalized clustering.

The only discrepency with Jin (2015) was for nPCA, where we found 590 errors and Jin found 600 errors. We did find 600 errors in an earlier trial run, so we tested whether nPCA is sensitive to initialization. We found that the 590-error result was stable across seeds, though a few seeds give higher error up to 600, which likely explains our contradiction. The earlier 600-error result is therefore consistent with a less favorable k-means initialization under this implementation. The main comparison uses a fixed seed for reproducibility, while the seed check shows the range of initialization sensitivity.

In the plot, each point represents a blog. Its vertical position is the SCORE ratio, and its color shows its reference political label. The labels are used only for coloring after SCORE is fitted. The two groups tend to occupy different ranges of ratio values, although there is some overlap and a few outliers. The horizontal axis shows row order in the dataset and has no substantive meaning. If rows are grouped by label, that ordering can make the color separation look stronger than it is.

Synthetic experiment design

The simulation generates undirected networks from a two-community DCBM with (n = 500) nodes and

$$P = \begin{pmatrix} 0.08 & 0.02 \\ 0.02 & 0.08 \end{pmatrix}.$$

Within-community edges are more likely than between-community edges. Node degree weights are generated from a lognormal distribution with dispersion fixed at $\sigma = 0.55$, matching the project's preliminary

simulation. The analysis fits SCORE, oPCA, and nPCA to 30 independently generated networks. All three methods are evaluated on the same network within each replication using misclassification rate and ARI.

The generator caps extreme degree weights so that all Bernoulli probabilities remain at most 0.99. This safeguard slightly truncates the upper tail of the requested lognormal distribution. It samples each unordered node pair once and mirrors the edge to create an undirected network.

In the illustrated network (seed 291), all three methods recover the two communities accurately. nPCA makes 3 errors (0.6% misclassification rate, 0.976 ARI), SCORE makes 9 errors (1.8%, 0.929), and oPCA makes 10 errors (2.0%, 0.921). The adjacency matrix, ordered by true labels for display, shows denser within-community blocks than between-community edges. This pattern is consistent with the simulation design, as because the network was generated with that block structure, it is not an independent validation of the method.

Across 30 networks at this fixed parameter setting, nPCA has the lowest mean misclassification rate (0.007, SD 0.004) and highest mean ARI (0.972, SD 0.016). oPCA has mean error 0.014 (SD 0.005) and mean ARI 0.944 (SD 0.019). SCORE has mean error 0.013 (SD 0.004) and mean ARI 0.947 (SD 0.017). Thus, all three methods recover these simulated communities rather well, with nPCA performing best on average in this particular design and SCORE close to oPCA.

The simulation ranking differs from the political blogs comparison, where SCORE performed best. This is not necessarily a contradiction: this simulated network follows one specific two-community DCBM with fixed block probabilities and a single degree-heterogeneity level. These results show that nPCA works especially well under these conditions, not that it is generally superior or that SCORE fails. Method rankings may change as degree heterogeneity, sparsity, and community separation change.

The analysis file `21_RMD_PreliminarySims.Rmd` contains the simulation code and plots. For the 30-replication pilot, the reported summary is:

Method	Mean error rate	SD error rate	Mean ARI	SD ARI
nPCA	0.007	0.004	0.972	0.016
oPCA	0.014	0.005	0.944	0.019
SCORE	0.013	0.004	0.947	0.017

Across 30 simulated networks, all three methods recovered the communities with high accuracy. nPCA performed best on average, with a mean error rate of 0.007 and mean ARI of 0.972. SCORE and oPCA had similar results, with SCORE slightly outperforming oPCA on both metrics. The standard deviations were modest, indicating fairly consistent performance across replications. These findings describe the current DCBM setting and do not establish a general ranking of the methods.

The ARI boxplot shows that nPCA had the strongest agreement with the true communities: its median ARI is highest, and most values are near 1. SCORE and oPCA had lower, fairly similar ARIs, with SCORE slightly higher overall. The error-rate boxplot tells the same story: nPCA usually made fewer classification errors. SCORE and oPCA had higher median errors, and oPCA's results varied somewhat more across simulations. All three methods performed well in this experiment. These plots summarize the specific simulated setting; they do not show how the ranking might change with different levels of degree heterogeneity, sparsity, or community separation.

Future Steps

Systematic simulation plan

The completed pilot uses one DCBM configuration: 500 nodes, two balanced communities, within-community probability 0.08, between-community probability 0.02, and lognormal degree heterogeneity $\sigma = 0.55$. The

next simulation stage will vary degree heterogeneity, community separation, and overall sparsity while holding the other parameters fixed within each comparison. Additional runs can vary network size and the number and balance of communities. If time permits, a robustness experiment will add edge-observation errors or dependence to assess how methods behave outside the model's conditional-independence assumption. Each setting will use repeated networks, with SCORE, oPCA, and nPCA applied to the same network within each replication. I will report mean and standard deviation of permutation-adjusted misclassification rate and ARI, with comparative boxplots. These metrics directly evaluate recovery of categorical community labels.

My project partner's assigned method is SIMPLE, which tests whether a selected pair of nodes has the same membership profile, returning a test statistic and p-value. This differs from SCORE's task of assigning one community label to every node. I will compare them on shared simulated networks using task-appropriate measures: node-level error and ARI for SCORE, and Type-I error, p-value calibration, and power for SIMPLE. On pure-membership simulations, I will also compare pairwise decisions: SCORE treats two nodes in the same estimated cluster as a same-community pair, while SIMPLE tests equality of the pair's membership profiles. If mixed-membership simulations are added, SIMPLE can be evaluated on profile equality, as SCORE's hard labels do not estimate a full membership profile. Analyses involving many node pairs will account for multiple testing. Runtime and sensitivity to random initialization may be compared as secondary measures.

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